CHAPTER

1

Human

Introduction

We are moving towards a world of automation with a goal to reach *The Era of Machines*. Each day our machines are remodeled with more features that can ease our task further. For example, washing clothes was a household chore. People found it cumbersome, so a machine was invented. These semi-automatic washing machines required human intervention, so fully automatic machines came into existence. Further these machines are enhanced each day to provide more features, like temperature, soak, dry, etc. Now we are trying to add intelligence to these machines, so that they can automatically set the wash time, decide on the amount of soap required, and so on. We are in a competitive world, and adding new functionalities is the only approach for a business to stay successful. In Figure 1.1 we depict the evolution of washing machines.

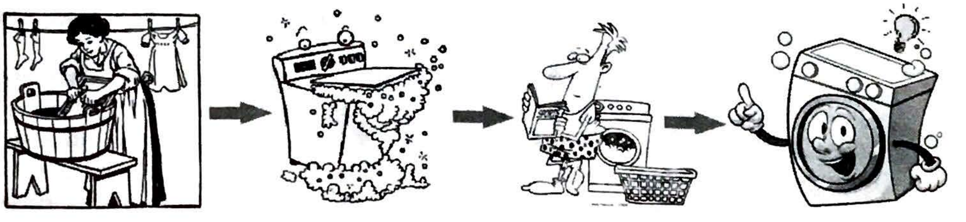


Figure 1.1 Evolution of washing clothes.

Imagine you have purchased a washing machine that is just launched. The promoting advertisements say that this machine can decide the amount of soap required, set its own temperature and timer. The advertisement even says there is no need of any kind of installation. You just need to connect the water pipes and you are good to wash. You make an online purchase, and this product has arrived at your door step. You open the wrapper, and find this machine really huge, as compared to your previous machine. The color given to its body is bright fuchsia with shiny gold rims. Your jaw drops a little. Even if pink were to be your favorite color, you may not prefer it on a home appliance. Now you place it in the destined location and want to connect the inlet and outlet pipes. The designer has concealed these valves so that the machine looks elegant. You will land up searching these valves for a while, reach out for the manual, and then discover them. The same thing happens with the power adaptor. Let us say, that you have managed to make these connections now.

Similarly the door has a concealed handle, to maintain the elegance. Again you read the manual and learn to open the door. Now you put a few clothes in it, and want to try the first wash. You are wondering how to start it. There is no display unit in the front of the machine, where it is supposed to be. You start searching for it, and again reach out for the manual. The designer has placed a power button on the side to avoid the accidental presses. You find this idea interesting, and now press on the start button, but nothing happens. Then you realize you have not added soap. From the manual you discover that the soap tray is placed on the side, next to the power button. Now, you drag the machine from its location, because the side wall does not allow you to open the soap tray. You add some soap and press the power again. The machine does nothing. You are anxious and wonder if the machine is defective. You read the manual over and over again, search the machine for more buttons, but nothing is visible anywhere. So you decide to call the customer care. After waiting on their line for more than twenty minutes, you get to speak to an executive. He says that the machine is fully automatic and you do not need to do anything more than that. You read the manual again, word by word. You check all your connections, the power source. You search for a display panel everywhere on the machine. You press the power button a few more times, and wait for a few hours. Nothing happens.

Explain eg of washing mc. with fully automization & reduction in interface.

What has gone wrong here? Is the machine really faulty? How could you decide that? Now everything about the machine will disturb you. The bright fuchsia body and shiny gold handles, the number of times you had read the manual, you had to move the machine just to add soap, the 20minutes wait on the customer care line, your search for the valves and soap tray, the invisible display unit, everything annoys you. Finally, you log into the online portal and book for a refund. This will leave the company with one unhappy customer, and lose thousands of potential customers who would read your reviews.

The millions of rupees spent on the research and innovation of all the features have become useless. Is the machine really faulty, or is the customer so dumb? Of course the machine works fine, but it did not have a good way of interacting with the user. The designer wanted to make the system simple to use, by eliminating all interactions. The machine is designed to start at the press of the button and the rest of the process is made automatic. It first calculates the temperature of water required, decides the amount of time and soap, and then goes into soak mode. But this was not known to you, so you pressed the button again. That shut down the system. Again you press the power button and wait to see no change, and press the button again, that stopped it.

An interface is the way in which the machine is presented to the human.

Each person has a different level of intelligence and experience. Human Machine Interaction is the field of research that concentrates on how to present the functionality of a system to the user. A huge amount of study on human psychology is required to build a good interface. Every single aspect about the target users' needs to be considered, such as their interests, behaviors, needs, likes, dislikes, experience, challenges, etc.

Technology may change rapidly, but people change slowly. Hence a research on human psychology will yield results that remain true forever. This book aims at bridging the gap between human psychology and the upcoming technologies, so that we design more human-centric models. As the saying "only a good listener can be a good speaker", in HMI we can say "only a good observer can be a good designer".

History of User Interface Designing

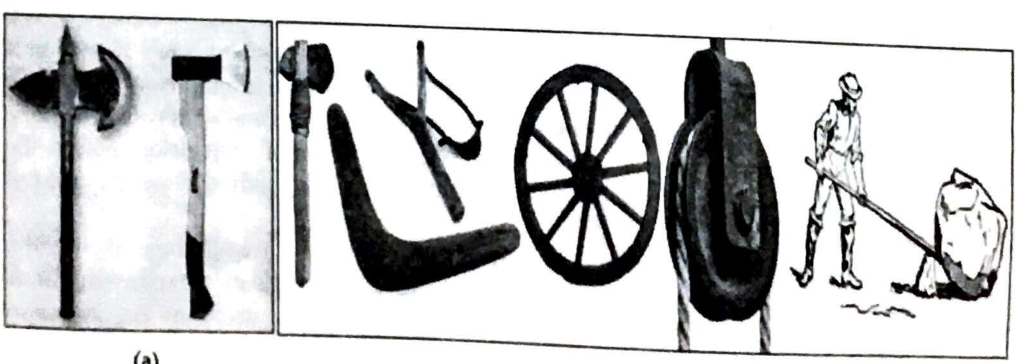
We begin our computer engineering and information technology courses with generations of computers. Similar to that user interfaces also have generations.

A machine can be defined as anything that can reduce manual effort.

First Generation: Machines that Reduce Physical Labor

The earliest machine that can be thought of is the hand-axe, a machine to transform lateral force into a transverse split. The handle is the interface to access this tool. A long wooden piece that is rounded smoothly is used to ensure a good grip. The shape, size, weight and texture of the wedge and the handle decide the outcome. The crude hand-made axes of Stone-age gradually improved in aesthetics as well as purpose. Look at the axes in Figure 1.2(a). You can easily identify the woodcutting axe from a battle axe, just by their appearance. Slowly more machines were invented, such as boomerangs,

Interaction medium was Shape, Size, material, used



(a) (b)

Figure 1.2 Machines to reduce manual labor.

slingshots, wheel and axle, pulley, levers, etc. For each of these machines, their physical appearance such as their shape, size, material used, weight, etc., decided their utility. Figure 1.2(b) shows a few machines that are used to reduce physical labor.

Second Generation: Machines that Displayed Output

These machines calculate the speed of wind, temperature, time, etc. They showed humans some kind of readings. The presentation of data to the user can be called as its interface. The units of measurements, the shape and size of pointers, color, scale, etc., made a difference to the ease of use of the machine.

The earliest known machine of this genre is a sundial. In the beginning, only a gnomon (a vertical pillar) was used. The length of its shadow indicated time. Slowly a dial was built onto which the gnomon was installed. The precision required to space the numbers on the dial determined the accuracy of time read. Gradually the dial included various data such as directions, zodiac constellations etc. Figure 1.3 shows a few machines that provide us with some readings of the environment.

Figure 1.3 Machines with output.

Third Generation: Machines that provided Output with Feedback

Feedback is an acknowledgement a user receives from the machine when his action is registered. Most of the home appliances like fan, lights, television, mixer, microwave oven, washing machine, coffee maker, air-conditioner etc., come under this category. For example, if user presses on the channel change button of a television remote, the user can see the channel change on the screen. User can immediately see the effect of his action.

Here the interface is the part through which the user is able to make the system perform some action such as switches, knobs, remote controls etc. The design of this interface is totally the deciding factor of the entire product's usability. We can see in our houses how each of our appliances has evolved over time, and also their usability. Electrical switches have become safer and softer to handle, mixer knobs are easier to operate now, TV remotes got standardized, etc. Most of us are familiar with most of such machines. We have operated an older version when we were children, and now we have modern ones. With a keen eye, you can observe how the interfaces of each of these devices have evolved over time. Figure 1.4 shows a few machines that provide feedback to users' actions.

Figure 1.4 Machines with feedback.

Fourth Generation: Machines with Computing Power

The innovation of computers has motivated research in various fields. Earlier providing functionality itself was a challenge, and was a field of research. Today we have options, and we are free to discard the ones we are not comfortable with. For example, earlier people learnt DOS commands to work with a computer. Today no one puts so much effort to learn to use a system. They just shift to a system which is easier to use. Human Computer Interaction is a field of research that emerged due to this competition. Here the focus is on easing the communication between the user and a computer.

Earlier only text-based screens were available. With the introduction of graphics, the scope of creativity in user interfaces has magnified. Technology is changing rapidly, and we have new computing devices every day. From the huge main frame computers we have shrunk to smart- watches, which have the same capabilities! Today's biggest challenge is to present all the functionalities in that limited display space and in the most user friendly way. Figure 1.5 shows the various computing devices we use. We can see the way in which their sizes have shrunk over time, but their functionalities have increased.

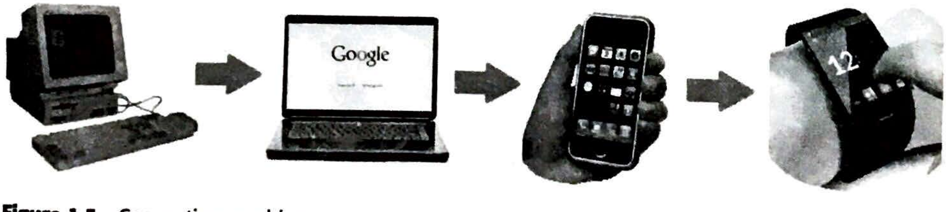


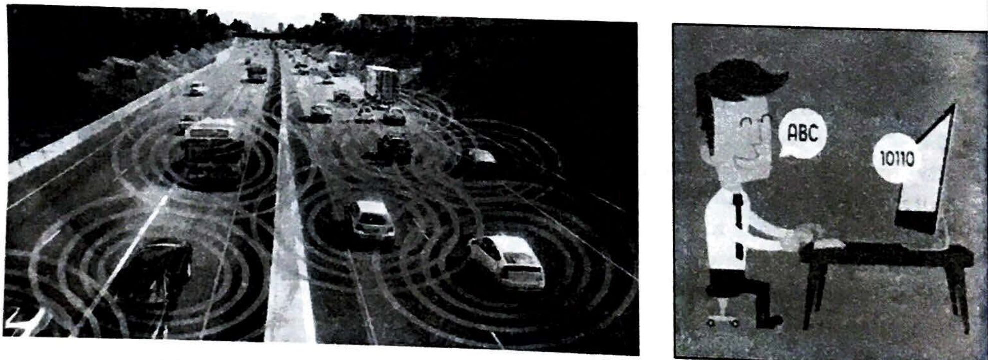
Figure 1.5 Computing machines.

Fifth Generation: Intelligent Machines

With artificial intelligence in boom, traditional input methods are proving to insufficient. We are making machines that can adapt to the humans by continuous learning. These systems require the freedom to learn from the environment. Here the user need not learn how to use the system. Instead these systems learn how the user is going to operate the system. These systems are capable of acquiring inputs independently, as well as learn by analyzing them. Most of the prediction systems, business intelligent systems, tracking systems, sensors, etc., are intelligent. With traditional input methods, these systems cannot provide the desired outcomes. Figure 1.6(a) shows automobiles with obstacle sensors. Input is taken directly from the environment, and is not typed into or pointed by any particular human.

Consider an automatic air conditioner. It scans the room to get number of people, fetches the outside temperature, and then automatically adjusts its temperature to the optimal value. For such systems keyboard, mouse or even a touchscreen restrict the learning of the machine. Natural language processing is an upcoming field that allows humans to freely talk to the computer in their own language, and the computer will interpret it correctly as depicted in Figure 1.6(b).

These systems also learn about humans from their actions. The restriction to hold a device in his hand and provide inputs in a specific way will not help the machine learn. With voice inputs, body gestures or brain signals, all such restrictions are removed, and there is a wider scope of machine learning. In voice UIs, the user communicates with the computer completely with his voice. Gesture UIs on the other hand take input from users' bodily movements. While the brain-computer UIs, can operate the systems with just the brain signals. Users need not give any external output in anyway.



(a) Obstacle sensors in automobiles

(b) Natural language processing

Figure 1.6 Intelligent machines.

Future: Augmented Reality

This concept is the opposite of virtual reality. You may have come across many games that involve building a new city, a farm or a house and family. This is virtual reality, game of building new city

where the real world is replaced by a virtual world. In augmented reality, the virtual world is brought into a user's reality. We watch many sci-fi movies where a human is able to enter into a video game, or one of the game characters come into our real world. We see video conferences with 3D holograms, sunglasses that able to pull out database records about the things in your line of sight, and reprogrammed humans. Such is the scope of augmented reality. Here not only the input methods should have freedom, even the output has to be unrestricted. You cannot travel into a user's reality by making him hold a 4 × 6 inch screen in his hand. The future of UIs is to make the line between reality and virtual world so blurred, that we can miss it easily. Figure 1.7 gives an idea of what kind of output is perceived by augmented reality.

eg User can go in game or any character from game come to u.

Figure 1.7 Augmented reality.

Hardware, Software and Operating Environments

Hardware

Hardware was the most important utility in earlier generations of computing. They were sole game players, as hardware use to drive the software. But in the current scenario, it is quite the opposite. Today we can choose the hardware as per the requirement of the user, and can top it up with any software application. The hardware options available are plenty, and one need not compromise like earlier times.

Software

Software is the tool with which we can create an effective user interface. Again we have a very large range of software that can be chosen. Based on the requirement of the task, and the underlying hardware, we need to decide on the software. It can be lower level languages, such as Assembly Level Languages, or high level languages, like C, C++, Java, and so on. There are many front-end developer tools, with which we can create an audio/visual experience for the user, such as Visual Basic, HTML5, PHP, animators, etc.

Operating Environment

Though there is huge turnover in the software and hardware, the major focus is on the end user and based on that the designs are decided. Our design decision should fulfill the user-level acceptance test and the modification should be provided immediately after any suggestion. The operation environment should follow a process within that certain level of pitfalls should be avoided.

Following are the some key points that define the golden rules of user interface design:

- Friends, family members, colleagues are not representatives of target users.
- User requirements should be understood by a team and not by an individual.
- Goal should be to minimize user difficulties.
- The hardware (device) and software balance should be maintained.

User Interface

User interface is not necessarily a computer or mobile screen. It is the medium through which a user sees the machine. It may be in the form of hard copies, computer screen operation environments, remote controls, or just the appearance of machine itself. A good UI will ensure that the target audience will be able to understand the functionality of the system by just seeing it. User requirements should be identified well in advance before designing the interface. In this chapter we focus mainly on the major factors about humans that need to be considered before the actual design process.

The Psychopathology of Everyday Things

The topic name is very intriguing for one. By breaking it down, word-by-word, it will be easier to understand the idea behind this topic. First, we will break the word *psychopathology* as:

psycho = mind; *pathos* = suffering/disease; *logy* = study.

Psychopathology: The study of mental illness

Psychopathology of everyday things: How the design of our everyday things can make us crazy.

Have you ever walked into a closed door? It is very frustrating to bang your head on the glass, and then realize that the door was meant to be pulled, and not pushed. This is a common experience, but it is not the fault of the user. Had the handles been designed in such a manner that it leaves enough clues for the human mind to recognize, this error would not occur. These are called Norman doors – A common term used to say that these doors are not comprehensible to the human mind. The door in Figure 1.8 is a PULL door. You can realize that by closely observing the hinges. But the absence of handles makes it seem like this is a PUSH door.

Figure 1.8 Norman Doors.

Complexity of Modern Devices

Every product is launched with something different. A business believes in thinking out of the box. Sometimes they are with enhanced features, or sometimes with just a new appearance. They make a namesake interface to make sure all their features are accessible to the user. The designers rarely make an effort to analyze the utility of their new design.

Sometimes, the designer just wants to make a difference, and creates existing products with a new appearance, such as the Norman pots or the coffee cup shown in Figure 1.9. The coffee cup sure has an interesting appearance, but the designer again has not considered the difficulty of the user to drink from it.

Technology is advancing at a great pace and the designs are not that competent. Focus of the developer is to equip the machines with more features, and exquisite appearances, but very less on the utility. A designer has to ensure that these features are presented in a simple and comprehensible way to the user, yet ensure perfect usability. Instead, they leave the user confused and make him feel incompetent to use such a high end product.

Most of us face trouble using simple day-to-day devices. To figure out the buttons on our TV or AC remote, we feel the need of an engineering degree. A new TV may be upgraded with many new features, but a user sees all these features through the remote that is in the user's hand. If the remote control confuses the user, then all the

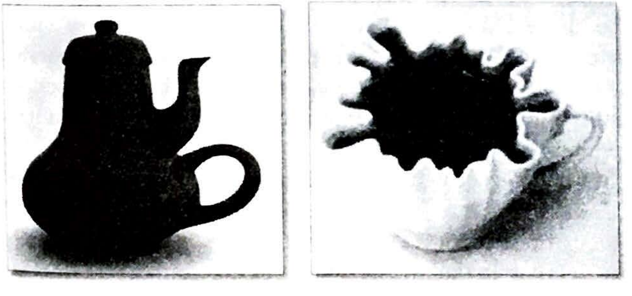


Figure 1.9 Confusing designs.

features provided become useless. Look at all the remote controls shown in Figure 1.10. Not all are easy to simple or easy to use by a novice user.

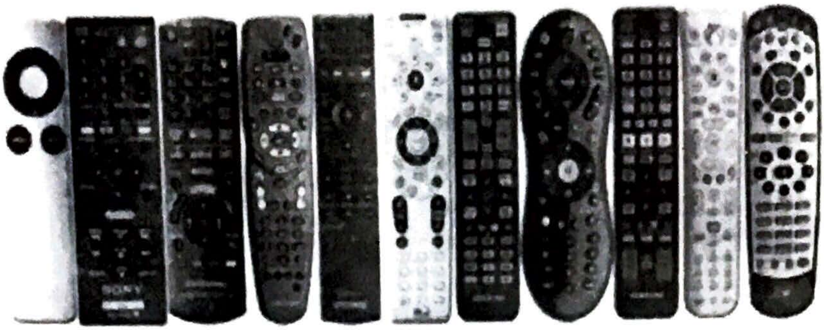


Figure 1.10 Remote controls.

Be it any appliance or software, technologies keep upgrading. Be it in microwaves ovens, air conditioners, washing machines, mixers, coffee makers, gas stoves, computers, mobile phones, wrist watches or treadmills, computing power is added. As the features increase, the complexity of the interface too grows. It is the responsibility of the designers to make these features available to the user in the most usable way.

Human-Centered Design

Human-centered designing is to consider all the aspects of the destined user, such as his interests, behaviors, needs, likes, dislikes, skill set, experience, challenges, etc., and build products that users will be able to easily adapt to. Analyzing the human characteristics is the requirement of the product design, and user satisfaction is the designer's goal.

Each person has some kind of experience of the real world, and their intelligence will use this experience to learn new features. It is the creativity of the designer to find this knowledge of the user, and exploit it to their advantage. For example, when you look at a suitcase, you know that it can be opened and you can place things inside it. Using the image of a suitcase on the computer screen will tell the user that it can be opened, and they will be able to find something inside it.

Conceptual model is the mental image a human has built. To be able to find things inside a suitcase is the mental image. Similarly, when you have to make a note of a few points, you look for a pen and paper. This is again the mental image formed for writing. As a good designer you should be able to use these mental models of the user, as well as be able to create mental models for the user. For example, Microsoft Word has become the mental model of a text editor unanimously.

Conceptual models define a good design as the communication between the designer and the user. The designer must be able to explain the entire product to the user by just the appearance of it. These models are very critical for a good user experience. If the end product does not map to the mental images of the user, the product is not a success.

Norman talks of four aspects through which these conceptual models can be used and/or constructed.

1. **Feedback:** The effect of every action. A feedback in any form is very critical to the user. In the earlier washing machine example, the user did not get any kind of feedback from the system. That made the user assume the system is faulty. Every single user action has to be acknowledged immediately. Giving acknowledgement to every user action.
2. **Constraints:** Prevent the user from making mistakes. Instead of having an option for the user to make a mistake and then forgiving them, make sure your user can never make a mistake. For example, you want your user to enter a date. Show a pick-n-click calendar, instead of a textbox. This will eliminate all possibilities of syntax mismatches (see Figure 1.11).

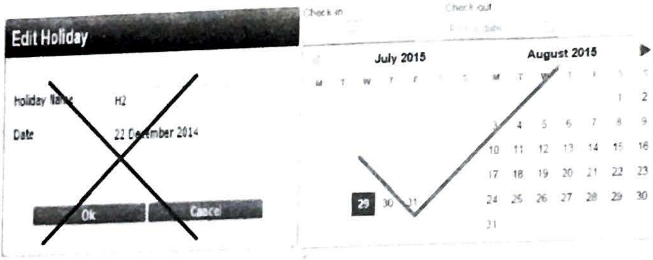


Figure 1.11 Constraints for date entry.

To leave messages everywhere, such as "Click here", "Insert this way", etc., are the qualities of a poor design.

3. **Affordances:** Convey the rules by leaving visual clues. To make sure that the appropriate actions are perceivable, and non-accessible ones are not invisible. By just the appearance of any object, its functionality must be clear to the user. Example, by looking at the handles, we should know how the door opens.
4. **Power of observation:** Learn from the struggle of others. As a normal person, we do not have a keen eye to observe. In fact when someone is making a mistake or struggling with a system, we tend to make fun of them or judge them to be incapable. As a HMI student, you need to observe them. Their struggle could be genuine and their mistakes could be innocent. As designers, we need to consider every such aspect of the user, and ensure that he/she gets a good experience.

Norman's Fundamental Principles of Interaction

Affordances

'what environment offers to the individual.

These are the little aspects that tell the human what needs to be done. This concept was introduced by psychologist J. J. Gibson. He defines affordances as "*the qualities of the physical world that suggest the possibility of interaction relative to the ability of an actor (person or animal) to interact*". He stated three fundamental qualities of an affordance:

- An affordance exists relative to the action capabilities of a particular actor.
- The existence of an affordance is independent of the actor's ability to perceive it (emphasis added).
- An affordance does not change as the needs and goals of the actor change.

You may notice many containers in your kitchen. Each opens in its own way. Not many have any instructions printed on them. They may open with a screw movement or by a pull/ press technique, while some have a locking feature. How are you able to identify one from the other? From the containers in Figure 1.12, are you able to guess the way each can be opened? Is it the shape or size of the body? Or is it the shape of the lid? Or do both participate? Those features of the physical appearance than contribute in explaining to the user the mechanism with which it operates can be called as its affordances. Here the shape of the lid and body are the affordances of the container.

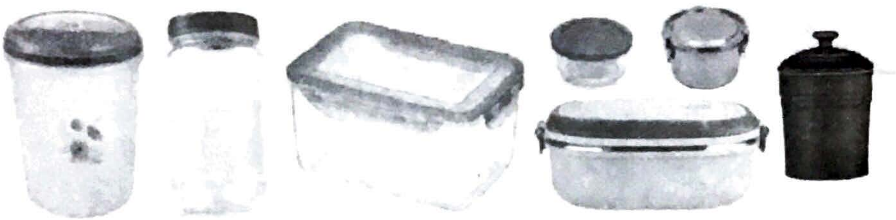


Figure 1.12 Kitchen containers.

For example, when you see the handle of a cup, you know that is provided to lift the cup. Similarly, when you see a 3D button on the screen, you tend to click on it. The blinking vertical line tells you that you can type here (Figure 1.13). Similarly, a round red button in a remote control tells the user that it can be pressed to power on or off the system. *Affordances are the visual clues that lead the user to understand the functionality of the object.* It is the most critical aspect of a design. As a designer, you should first realize the users' level of experience and intelligence. Only then you can make a design that would be predictable to the user.

Figure 1.13 Textbox with blinking cursor.

Signifiers

This is a physical form of showing the functionality to the user, such as a sound, a printed word, or an image. For example, writing the word "PUSH" on a door is a clear way to tell the user that the door will open when pushed. Sometimes, affordances are not clear enough to tell the user the functionality of the object. A signifier removes all such ambiguity. In an ATM machine the card insert slot can be confusing at times. You may wonder whether you need to swipe the card and remove it, or let the machine absorb the card. In such cases, you may add a signifier INSERT or SWIPE (Figure 1.14).

Figure 1.14 Signifiers.

Affordances, Perceived Affordances and Signifiers

Perceived affordance is what the user understands by looking at the object. It may not be the same as what the designer intended it to be. This concept was introduced by Norman. Sometimes we look at an object and assume its purpose. In Figure 1.15(a) the panel of a lift is shown. Here it is easy for the user to assume that the glowing arrows are pushable. Only after pushing it the user will realize that it is just a label, and it cannot be pushed. After that realization starts the next confusion. To go down, should you press the square beside the down arrow, or the one beneath it?

Another example, many novice users look at the two *Enter* keys on a keyboard as shown in Figure 1.15(b), and assume they have different functionalities. The intention of the designer was to increase the usability. Since most of the keyboards have these two *Enter* keys in different shapes, many novice users easily form this assumption. Sometimes you may even hear them say "Press this ENTER key, not that one!"

(a) Lift panel

(b) Keyboard with two Enter keys

Figure 1.15 Perceived affordances.

Mapping

To present the relationship between two objects is mapping. Here we shall map actions to consequences. Every user action has to be acknowledged in some way. In the beginning of the chapter we saw how a poor feedback system can frustrate the user, to the extent of returning his brand new washing machine. Had his actions been acknowledged in some way, his anxiety will reduce and will have a happy customer.

It is also important to have a good mapping between the action and consequence. The user must be able to predict the consequence of his action beforehand. If the user moves his mouse to the right, the pointer on the screen also moves to the right. It is easy for the user to learn this mapping and remember it too, because the movement of the hand directly affects the movement of the pointer. They work synchronously. Such mapping can be termed as natural.

Steering a car wheel in the clockwise direction, turns the car to the right. Most of us are trained to view the clockwise rotation is a rightward movement, hence this too becomes a natural mapping. Similarly to increase the volume of a device, we turn the knob clockwise. It is imbibed in most humans to assume an increase as we go rightwards. Could be because of the mathematics teacher in our schools who taught us the greater than symbol "<".

Sometimes the fan regulators get fixed the opposite way (Figure 1.16). It may be due to the negligence of a careless electrician. Even after months or years of its usage, you will still try to rotate it the opposite way, and curse the electrician for his poor job.

Figure 1.16 Fan regulator.

Switches provide immediate feedback to the user about his action. You push a switch, a light glows. Imagine you are sitting in a room of switches, all of whose light bulbs are in other rooms. How long will it take for you to learn the mapping of the switches to their correct bulbs? You take time because you are not able to see the consequence of your action. You need to train your brain systematically with repeated usage to remember this mapping. It is not a natural process. Only when a user can see the consequence of his action, he will learn and remember its functionality.

Exploit the physical analogies and traditions of your user to create the best products. It is easier for us to operate a machine with the knowledge we already possess than to learn and remember something new. Use natural mapping as much as possible.

Feedback

Feedback is the best and the fastest way of learning. We humans always work on a trial-and-error method. We want to try to use the product first and learn it during the course. You do not want to theoretically study and earn a degree on how to use your mobile phone before purchasing it. You go to the store and pick the latest model that is affordable for you. You believe you will be able to learn its working as you start using it. Unlike old phones, these days mobile phones come with a lot of features on a small display screen. Not all icons provided are comprehensible of their exact use. It takes time for one to learn all these functionalities.

Feedback should not only be prompt, but also predictable to the user. A correct mapping of the actions and consequences creates a good feedback system.

Conceptual Models

The affordances induce a thought of action. Signifiers affirm that thought. Mapping his action to a consequence provides a feedback. Good feedback encourages learning. Repeated learning creates a mental model in the user about the system.

"I think the system does this" is different from what the system actually does. A conceptual model is developed gradually in the user's mind after repeated interactions with a system. A design model is the conceptual model of the designer, and not the user. They can be very different from each other if the system is not well designed.

A driver's mental model says that the press of the accelerator pedal of a car makes the car go faster. The design model is to provide more fuel to the engine when the accelerator is pressed. Here the mental pictures may be different, but the consequence is the same. Hence there is no problem.

Here is a customer care center joke that was commonly shared a few years back.

Executive: How may I help you?
Customer: The coffee cup holder of my PC broke this morning, and I'm still in warranty.
Executive: I'm sorry I do not understand. Was this cup holder given to you as a compliment with our computer?
Customer: No. It is inside your computer. I press this button, and it pops out.
Executive: But we do not have any computers that come with cup holders. Is there anything written on this button?
Customer: Yes. HD DVD

You may laugh at this joke. But there were many who really used the CD ROM tray as their coffee cup holders. A novice user does not know anything about computers or discs, but is familiar with the concept of placing coffee cups in round slots. He finds a tray with a slot that easily fits his cup, and in the ideal height to place his coffee while working on the computer (Figure 1.17). He in fact compliments the designers for their thoughtfulness. Can you really laugh at these users? Here the mental model of the user is a coffee cup holder, whereas the design was to insert discs.

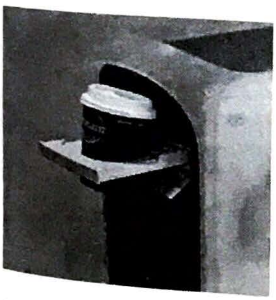


Figure 1.17 Cup Holder Tray.

It is the responsibility of the designer to first understand the mental model of users, and try to make a product that will agree to it. To make a user learn a new concept, and develop a mental model about the product will be difficult, as well as time consuming. Creativity for a designer is essential, but not at the cost of making the users feel incompetent.

Conceptual models is a vast concept that is covered in detail later in this book. This section should give you a basic idea of its significance.

Psychology of Everyday Actions

How do People do Things

Humans' first response to a poor design is self-blame. We feel incompetent to use the product. We first try the trial-and-error method. When that fails, we ask people around us to help us. When they too give up, we reach out to the user manual. Even after this if the product is incomprehensible, we call the customer care. We make every effort to learn about the product, because humans have the tendency to believe that the new product is beyond their level of expertise and they have to make an effort to understand it.

Also, humans also have a tendency to compare themselves. If the product has a good market value, but they are not able to figure it out, then the human feels incompetent to use the product. Only after we have tried everything under the sun, and the market reviews are bad, we tend to blame the product.

As a designer we can exploit this nature of humans. We can build high-end systems with complex designs and ask our end users to adapt themselves to it, and we can be sure that they will make every effort to do it as well. The disadvantage of this approach is that, humans can get exhausted trying to adapt to new technologies. At one point, they will settle down for something known and easy to use, over a complex system that they need to learn all over again.

Our aim should not be to make humans adapt to our product.
Instead to build products that can adapt to the humans.

When we look at a product, we should know how to use it without any external help. Only then we can say that the product is good. We may be able to use it out of our experience or our intelligence. Either way, we can call that design successful.

Seven Stages of Action and Three Levels of Processing

Seven Stages of Action

Norman states that human actions have two aspects, execution and evaluation. We perform an action (execution) and then analyze our action if that is what we wanted (evaluation). Every action is performed in seven stages as given below:

1. Forming the goal

Execution

2. Forming the intention (plan)

3. Specifying an action (specify)
4. Executing the action (perform)

Evaluation

5. Perceiving the state of the world (perceive)
6. Interpreting the state of the world (reflect)
7. Evaluating the outcome (compare)

To understand this concept, let us take a macro example. You are alone at home and bored, so you go and watch a movie. Though this looks simple, our human brain executed it in the following way:

1. I want to kill my boredom (goal)
2. Movie seems to be a good idea (plan)
3. Check for the nearest cinema and show time (specify)
4. Purchase a ticket and go sit in the movie hall. (perform)
5. You watch the audio visual effects of the movie (perceive)
6. You interpret the effects to your understanding (reflect)
7. After the movie, you say to yourself "that was a good time pass" (compare)

Similar stages take place even in our micro activities. Let us take the example of a single action. You are driving a nail into a piece of furniture.

1. This nail has to hold these two blocks together (goal)
2. This location seems perfect (plan)
3. Hold the nail at that point (specify)
4. Start hammering it (perform)
5. I can see the hammer striking the nail, and hear the sound (perceive)
6. This nail seems to be going in straight (reflect)
7. At the end, I move the blocks to check if the nail has fastened them together (compare)

Though the action may look simple, most of the time we go through all these stages to perform it. There is always a gap between what we wanted to do, and what we have done. Norman calls this as *The Gulfs of Execution and Evaluation*. In the first case, what you wanted was to kill boredom, and what had done was watched a movie. In the second case, you wanted to fasten two wooden blocks together, and hammered a nail into it. Most of the time, what we do does not satisfy our goal. We can use this human feature in interface designing too. Since we know how exactly the user is going to perform every action, we can decide the flow for him.

- **Gulf of Execution:** The distance between the options available and the user's goal is the gulf of execution. You need to find if the system provides actions that correspond to the intentions of the person.
- **Gulf of Evaluation:** The amount of effort a person has to put to interpret the options available on the system and determine if they will match his intentions.

These seven stages of action are very much effective in deciding the user's perspective of the interface. If end user expectations are not properly analyzed while performing some actions users will

(a) Fail at few points out of which they may try to represent some different action
(b) User interface will not be user friendly
(c) User actions may be scattered
(d) User may end up with non-satisfactory results.

Three Levels of Processing

- **Visceral level:** The most immediate level of processing. Here the human reacts to audio, visual and other aspects of a product before experiencing it. The look and feel of the product dominates the user in this level. An immediate decision is made about what is good, bad, safe or dangerous. Bright colors, sweet taste, melody, good aroma, smooth surface, etc., are appreciated in this level. Our brain is trained to like or dislike based on our cultural beliefs and social upbringing. Girls like pink; boys like blue; Indians like spicy food; Brazilians like Samba; English like tea; etc. A taste needs to be developed to like bitter gourd or beer. As the saying, "beauty is in the eyes of the beholder", each person finds something more attractive than the other.

  Visceral design often corresponds to creating an aesthetically pleasing appearance. The creative skills of a visual and graphic designer alone creates an impact on the user. Only after this level approves of the product, we proceed towards working with it. If, at this level, we react with a negative feeling, the other levels will need tremendous effort to accept the product over long periods of time.

  The designer of a product with high end features may find it absurd to choose a good color for the product. But for the user, this becomes one of the most critical deciding factors.

- **Behavioral level:** The middle level of processing. The emotional brain takes control of the decision making. We react to the products at a deeper level than that of visceral. We get annoyed or frustrated with complex systems. We enjoy working on familiar and simple environments. This level manages our simple, everyday actions. Here functionality of the product takes prime importance.

  In behavioral design, semantics, and usability practices are primarily addressed. It tells the user how to "behave" or "respond" to a communication or feedback given by the product. For example, a dialog box with an error or warning message tells the user what needs to be done next. Or a simple ON and OFF mark on an electrical switch tells us how to operate the switch.

- **Reflective level:** The last level of processing. A careful analysis and reflection of all the incidents or experiences is made. The meaning of an experience is stored in our brain.

  Through reflection, we try to share our experiences by designing models and convert them into prototype. Reflective design is nothing but the effort to develop long-term relationships with the customers. Thus, reflective design can tell us about the customer's inclination towards various ranges of products.

Knowing the three levels at which we process information (Figure 1.18), the seven stages of our actions can be understood better. Each stage of action works on a particular level. Through Figure 1.19, Norman gives an overview of this mapping.

At the reflective level a need is identified which becomes our goal. Analyzing our past experiences or future requirements, we make a plan to achieve this goal. Then based on our mood, likes and dislikes we choose a method to execute that plan. The behavioral level is responsible for this action. Most of the time our decisions are affected by our emotions. This level has the strongest influence over a human being. At the visceral level we actually perform the specified method. All the physical aspects of performing the task are considered. Our senses of vision, hearing, smell, touch and taste are used to perceive the action being performed. These perceptions are then interpreted and given a meaning by the behavioral level again. Finally the reflective level analyses our experience and compares it with the initial goal. It stores this experience as positive or negative for future reference.

Let us consider the same example of watching a movie because we were bored. The stages of action that were followed to execute it were seen in the earlier section.

1. **Goal:** I want to kill my boredom. The reflective level has analyzed your situation as boredom, and realizes from past experience that it should be avoided.
2. **Plan:** Movie seems to be a good idea. Again the reflective layer gathers from past experiences that a movie is a good distraction.

LOOK & FEEL	➔ ➔	Visceral level	➔ ➔	MOTOR SKILLS
	➔ ➔	Behavioral level	➔ ➔	
	➔ ➔	Reflective design	➔ ➔	

Figure 1.18 Three levels of processing.

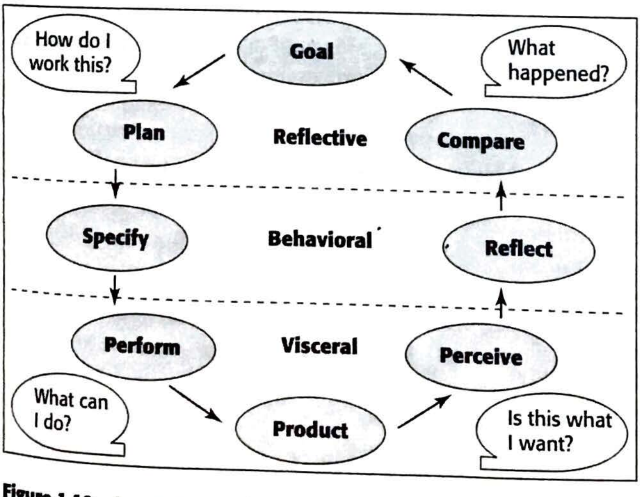


Figure 1.19 Seven stages of action and three levels of processing.

3. **Specify:** Check for the nearest cinema and show time. Now the behavioral layer has its likes and dislikes. It takes the responsibility of choosing the perfect movie for the moment.
4. **Perform:** Purchase a ticket and go sit in the movie hall. The visceral layer takes care of the physical activities, like choosing the seat, making payment, reaching the movie hall and watching the movie.
5. **Perceive:** You watch the audio visual effects of the movie. The visceral level appreciates the various audio and visual effects of the movie. The actions presented, the sound effects, the direction, etc., creates an impression about the movie.
6. **Reflect:** You interpret the effects to your understanding. Now the behavioral level reacts to these perceptions with an emotion. We relate ourselves with one of the actors emotionally. We cry, laugh and get angry while watching the movie. This level appreciates the story line of the movie.
7. **Compare:** After the movie, you say to yourself "that was a good time pass". Now the reflective level gives a meaning to the entire experience, by comparing it with the goal. It sure killed the boredom. So it was a good idea. The next time you feel bored you will consider watching another movie.

Summary

Through this chapter we introduce

- The aim of this field. The, aim to not make humans adapt to a product. Instead to build products that can adapt to the humans.
- Evolution of user interfaces by classifying machines into five generations.
- A basic understanding of all the problems faced by humans with day-to-day devices, and the need for a designer to look into these issues.
- Norman's principles of interactions – Affordances, Signifiers, Perceived Signifiers, Mapping, Feedback & Conceptual Model.
- A realization on human psychology that reveals how people think and perform every action – seven stages of action and three levels of processing.

Multiple Choice Questions

1. Interface is
   (a) GUI
   (b) Interaction method
   (c) Physical appearance
   (d) All of these
2. Data mining requires an interface that is of
   (a) Fourth generation machines
   (b) Fifth generation machines
   (c) Augmented reality
   (d) Backend processes do not require an interface.
3. Interfaces in augmented reality focus only on output.
   (a) True (b) False
4. The image of a horn placed on top of the button in a car is a

(a) Affordance
(b) Signifier
(c) Perceived signifier
(d) Feedback

5. We are able to differentiate between a battle axe and a woodcutter's axe because of their
(a) Affordances
(b) Signifiers
(c) Decoration
(d) Place of display

6. Revamping the appearance of an existing device makes it
(a) More user friendly
(b) Business competent
(c) Complex to use
(d) Annoying to the user

7. Natural mapping is relationship between
(a) Mental model and design model
(b) Cognitive skills and consequences
(c) User and machine
(d) Nature and culture

8. Norman doors are
(a) Doors without handles
(b) Revolving doors, as in a mall
(c) Confusing doors
(d) Invisible glass doors

9. Affordances work on
(a) Visceral level
(b) Behavioral level
(c) Reflective level
(d) All of these.

10. Match the following:

(a) Visceral level	(i) Emotional
(b) Behavioral level	(ii) Experience
(c) Reflective level	(iii) Reactive
(d) Conceptual model	(iv) Analytical

Review Questions

1. Explain the history of user interfaces.
2. How does the complexity of modern devices add to the psychopathology of these things?
3. What is a human-centric design?
4. Give Norman's fundamental principles of interaction.
5. Compare affordances and perceived affordances.
6. What is the difference between mapping and feedback?
7. How does feedback help in developing mental models?
8. Why do humans blame themselves for a poorly designed product?
9. Explain the seven stages of action and three levels of processing with suitable examples.
10. Explain the gulf of execution and evaluation with an example.
11. Sit next to an elderly person working with a computer. Observe the mistakes they are making. Note down their reactions. Try to find out their mental image of the system in use. Identify at what level of processing they are facing the problems.
12. Pick an appliance in your house that is used the most. Observe all the options given on it, and try to create the design model of this product.
13. A light bulb is faulty in your house and you need to replace it with a new one. Describe the seven stages of action that you will follow to change this bulb, along with the level of processing in each.